

Semantics at an Angle: When Cosine Similarity Works Until It Doesn't

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Abstract

Cosine similarity is a standard comparison rule for learned representations in information retrieval, natural language processing, computer vision, and multimodal learning. Its popularity is well founded: it removes positive radial scale, is computationally convenient, and often matches objectives that train normalized embeddings. These same properties also delimit what cosine can express. Normalization discards radial variation; anisotropic representations can compress angular contrast; high-dimensional neighborhoods can develop hubs; and a symmetric, uncalibrated score may be mismatched to the relation of interest. This article offers a selective review organized around a simple principle: the usefulness of cosine similarity depends jointly on the learned representation, any normalization or post-processing, the scoring rule, and the downstream decision. We derive the main geometric identities, distinguish failure mechanisms that are often conflated, review representative evidence about embedding norms, and describe objective-matched, geometry-aware, hubness-aware, norm-aware, and learned alternatives. The central conclusion is conditional rather than adversarial: cosine's positive-scale invariance is justified when radial variation is nuisance or fixed by the representation contract, but its angular geometry and downstream decision must still be validated.

Keywords: cosine similarity; embeddings; representation geometry; vector norms; anisotropy; hubness; semantic similarity; retrieval

1 Introduction

Learned embeddings turn objects into vectors so that a downstream system can compare, retrieve, cluster, or classify them. Once an encoder $E : \mathcal{X} \rightarrow \mathbb{R}^d$ has been trained, it is tempting to regard the choice of comparison rule as a minor implementation detail. In practice, that choice imposes an additional modeling assumption. For nonzero vectors $\mathbf{x} = E(x)$ and $\mathbf{y} = E(y)$, cosine similarity retains only their directions. It therefore treats $\mathbf{x}$ and $a\mathbf{x}$ as equivalent for every $a > 0$, regardless of whether their different lengths arose from nuisance scale, confidence, frequency, input complexity, or task-relevant signal.

This observation does not imply that cosine similarity is generally defective. In sparse document retrieval, scale invariance attenuates document-length effects. In contrastive learning, embeddings are often normalized within the training objective, making a spherical comparison part of the model itself [1–3]. For any unit-normalized representation, cosine similarity, inner product, and Euclidean distance induce the same nearest-neighbor ranking. Under those conditions, cosine is not merely a convenient default; it is an objective-matched score.

The surrounding embedding vocabulary has also become substantially more visible. Figure 1 provides descriptive context by comparing search attention and publication counts for cosine similarity, text embedding, and retrieval-augmented generation. These curves measure attention to specific query strings, not technical merit or comparative accuracy.

Problems arise when this model contract is absent or misunderstood. The norm of a learned vector can contain information in some models [4, 5]; the directions of contextual representations can be strongly anisotropic [6]; and nearest-neighbor graphs can exhibit hubness even when all points have unit norm [7, 8]. These are distinct phenom-

ena. Recovering norms cannot by itself repair anisotropy or hubness, and making a representation isotropic does not make a symmetric score suitable for an asymmetric relation such as entailment.

The aim of this article is to make those distinctions precise while retaining a compact, tutorial style. It is a selective review rather than an exhaustive survey or systematic evidence synthesis: sources are selected to illustrate mechanisms and remedies, not to estimate an average effect across models. We emphasize foundational work and representative evidence available through 2025, with examples from text, vision, multimodal learning, and dense retrieval. Our contributions are fourfold:

- We state the geometric properties of cosine similarity carefully, including its invariances, its relationship to metric distances on the sphere, and the role of vector norm in optimization.
- We explain historically why cosine became a default while separating the training objective from the downstream evaluation score.
- We organize limitations into four operational diagnoses: radial information loss, global geometric distortion (including anisotropy and concentration), local-density or hubness effects, and task–score mismatch. We treat representation–score non-identifiability as a cross-cutting warning about interpreting learned coordinates.
- We give a practical decision procedure for choosing among cosine, inner product, post-processing, hubness correction, norm-aware scoring, and learned comparison functions.

The resulting thesis is deliberately conditional: *cosine similarity quotients out positive radial scale. That invariance is justified when radial scale is nuisance or fixed by the representation contract; it is not by itself sufficient to make cosine the right score for the downstream decision.*

2 Four Objects That Should Not Be Conflated

Claims that cosine works or fails are incomplete unless they specify what produced the vectors and how the scores are used. A useful analysis separates four objects.

Table 1 is the organizing frame used throughout this review. For example, anisotropy is primarily a property of a representation distribution, although post-processing can change it. Hubness is a prop-

erty of the induced neighborhood graph, not an attribute stored in the norm of one vector. Calibration is a property of scores relative to outcomes and a target population. Cosine normalization, in contrast, is a deterministic transformation followed by an inner product. Keeping these levels separate prevents a single angle-versus-norm story from being asked to explain every failure.

2.1 A cross-cutting identifiability warning

Even without normalization, the geometry of learned factors need not be uniquely identified by the data-fit term. Consider an unconstrained two-factor model whose predictions depend only on $\mathbf{X}\mathbf{Y}^\top$. For any invertible matrix $\mathbf{A}$,

$$(\mathbf{X}\mathbf{A})(\mathbf{Y}\mathbf{A}^{-\top})^\top = \mathbf{X}\mathbf{Y}^\top. \quad (1)$$

Equation (1) preserves this product and hence the model’s predictions, but cosines within the transformed factor spaces generally change unless $\mathbf{A}$ is a scaled orthogonal transformation. The full invariance need not survive tied parameterizations, normalization, explicit geometric constraints, or regularization; such choices can restrict or select among equivalent factorizations. Steck et al. [9] analyze how regularization controls this issue in linear embedding models. The applicable lesson is conditional: semantic interpretation requires checking the invariances of the actual training objective, not only the coordinates it outputs.

Terminology. We use *representation contract* for the combination of exposed representation, normalization, and pairwise interaction that a downstream user is expected to apply. A *unit-normalized interface* is one such contract. Throughout, *spherical* means that the compared embeddings have fixed norm (unit norm unless stated otherwise). It does not imply *isotropy*, which is a distributional property: ideally, a random direction $\mathbf{U}$ is isotropic if $\mathbf{Q}\mathbf{U} \stackrel{d}{=} \mathbf{U}$ for every orthogonal matrix $\mathbf{Q}$. Empirical anisotropy diagnostics measure departures from this ideal.

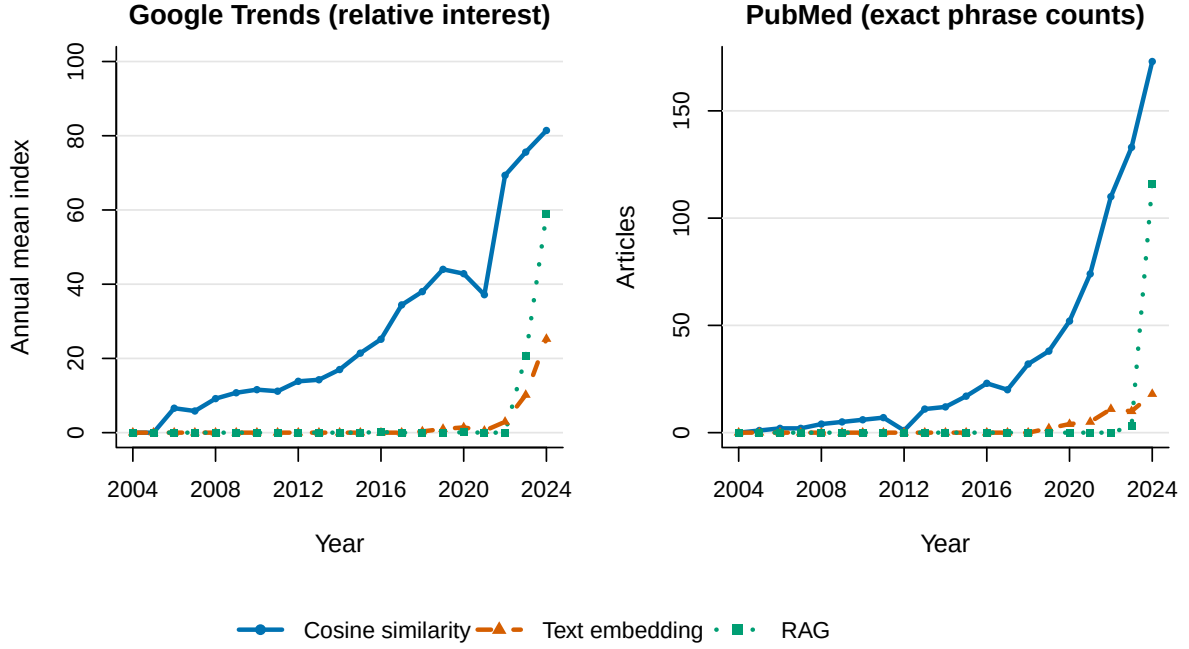


Figure 1: Descriptive attention to three embedding-related search terms, 2004–2024. Left: annual means from one joint worldwide Google Trends comparison of quoted phrases (relative 0–100 interest). Right: yearly PubMed counts for the exact phrases in Title/Abstract under the Publication Date filter, refreshed July 23, 2026. Trends values are sampled and normalized, so zeros may reflect low volume. The series measure attention to query strings, not method quality; acquisition details accompany the source.

Table 1: Four levels in an embedding comparison pipeline. A failure observed at one level should not automatically be attributed to another.

Level	Mathematical object	Question
Representation	$E(x) \in \mathbb{R}^d$	What information and invariances did the training objective encode?
Transformation	$T(E(x))$, such as normalization, centering, or whitening	Which variation is removed, rescaled, or estimated from a reference corpus?
Pairwise score	$s(T(E(x)), T(E(y)))$	Does the comparison use direction, norm, local density, or learned interactions?
Decision	ranking, threshold, clustering, or calibrated probability	What task loss is optimized, and how stable is the rule across domains?

3 Mathematical Anatomy

3.1 Direction, radius, and invariance

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d \setminus \{\mathbf{0}\}$. Write each vector in polar form,

$$\begin{aligned} \mathbf{x} &= r_x \mathbf{u}_x, \\ r_x &= \|\mathbf{x}\| > 0, \\ \mathbf{u}_x &= \frac{\mathbf{x}}{\|\mathbf{x}\|} \in \mathbb{S}^{d-1}. \end{aligned} \quad (2)$$

and analogously for $\mathbf{y}$. Cosine similarity is

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} = \mathbf{u}_x^\top \mathbf{u}_y = \cos \theta, \quad (3)$$

where $\theta \in [0, \pi]$ is the angle between the vectors. It is undefined when either vector is zero and takes values in $[-1, 1]$.

For $a, b > 0$,

$$\cos(a\mathbf{x}, b\mathbf{y}) = \cos(\mathbf{x}, \mathbf{y}). \quad (4)$$

Thus cosine is invariant to *positive* rescaling. Negative rescaling flips direction and, consequently, the sign of the score. Geometrically, normalization maps each positive ray in $\mathbb{R}^d \setminus \{\mathbf{0}\}$ to one point on $\mathbb{S}^{d-1}$. Every value carried only by r_x is removed from the downstream score.

3.2 Cosine is a similarity, not a metric

Cosine similarity is not a distance metric. The commonly used cosine dissimilarity

$$d_{\cos}(\mathbf{x}, \mathbf{y}) = 1 - \cos(\mathbf{x}, \mathbf{y}) \quad (5)$$

is nonnegative and symmetric. On $\mathbb{R}^d \setminus \{\mathbf{0}\}$, however, it assigns zero to distinct vectors on the same positive ray, and it can violate the triangle inequality. Restricting the domain to the unit sphere repairs identity of indiscernibles but not the triangle inequality. Two closely related quantities do define metrics on the unit sphere:

$$d_{\text{ang}}(\mathbf{x}, \mathbf{y}) = \arccos(\cos(\mathbf{x}, \mathbf{y})), \quad (6)$$

$$d_{\text{chord}}(\mathbf{x}, \mathbf{y}) = \|\mathbf{u}_x - \mathbf{u}_y\| = \sqrt{2 - 2\cos(\mathbf{x}, \mathbf{y})}. \quad (7)$$

The first is the geodesic angular distance and the second is the Euclidean chord distance. Both are strictly decreasing transformations of cosine similarity and therefore reverse, but otherwise preserve, pairwise rankings. The von Mises–Fisher family provides a probabilistic model whose log density depends on an inner product with a mean direction [10], but this does not make raw cosine scores calibrated probabilities.

3.3 Equivalence after normalization

The squared Euclidean distance decomposes as

$$\|\mathbf{x} - \mathbf{y}\|^2 = (r_x - r_y)^2 + 2r_x r_y \{1 - \cos(\mathbf{x}, \mathbf{y})\}. \quad (8)$$

When $r_x = r_y = 1$,

$$\begin{aligned} \mathbf{x}^\top \mathbf{y} &= \cos(\mathbf{x}, \mathbf{y}), \\ \|\mathbf{x} - \mathbf{y}\|^2 &= 2 - 2\cos(\mathbf{x}, \mathbf{y}). \end{aligned} \quad (9)$$

Consequently, cosine similarity, dot product, and Euclidean distance produce identical rankings over a collection of unit vectors. They may still yield different numerical thresholds or calibration maps, so ranking equivalence should not be confused with score equivalence.

Figure 2 summarizes these relationships. Cosine fixes only the angle θ ; ordinary Euclidean distance additionally depends on r_x and r_y , while normalization replaces the original endpoints by $\mathbf{u}_x$ and $\mathbf{u}_y$ and makes their Euclidean separation a chord of the unit sphere.

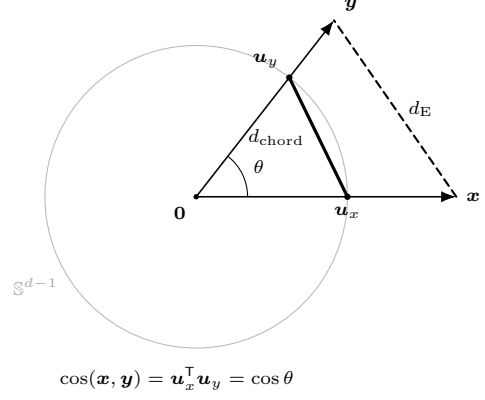


Figure 2: Cosine, chordal distance, and Euclidean distance in the plane spanned by $\mathbf{x} = r_x \mathbf{u}_x$ and $\mathbf{y} = r_y \mathbf{u}_y$. On the unit sphere, $d_{\text{chord}} = \|\mathbf{u}_x - \mathbf{u}_y\| = \sqrt{2 - 2\cos \theta}$. For the original vectors, $d_E^2 = \|\mathbf{x} - \mathbf{y}\|^2 = (r_x - r_y)^2 + 2r_x r_y (1 - \cos \theta)$. Thus Euclidean distance also depends on the radii; for unit vectors, it equals the chordal distance.

Table 2 makes the domain restrictions and retained information of the main comparison rules explicit.

3.4 Norms can affect learning even when the score ignores them

Cosine similarity is radially invariant as a function value, but its gradient with respect to an unnormalized embedding depends on that embedding’s norm:

$$\nabla_{\mathbf{x}} \cos(\mathbf{x}, \mathbf{y}) = \frac{1}{\|\mathbf{x}\|} \left(\mathbf{I} - \mathbf{u}_x \mathbf{u}_x^\top \right) \mathbf{u}_y. \quad (10)$$

The gradient is tangent to the sphere because it is orthogonal to $\mathbf{x}$, and its magnitude is $\|\nabla_{\mathbf{x}} \cos(\mathbf{x}, \mathbf{y})\| = \sin \theta / \|\mathbf{x}\|$. Thus it has inverse-radius scaling at a fixed nondegenerate angle but vanishes at exact alignment or anti-alignment. Pre-normalization radii can therefore affect optimization speed even though they do not affect the forward cosine score. Recent analysis of self-supervised learning makes this interaction explicit and also reports model-dependent relationships between pre-normalization norm and confidence [5]. This distinction between *the vector scored downstream* and *the vector that generated the training gradient* is essential.

4 How Cosine Became a Default

4.1 Sparse information retrieval

The SMART system was already being used to evaluate automatic indexing and retrieval procedures

Table 2: Common comparison rules. Metric refers to the mathematical distance axioms on the stated domain.

Rule	Definition	Uses norm?	Metric?	Key property
Cosine similarity	$\mathbf{x}^\top \mathbf{y} / (r_x r_y)$	No	No	Invariant to positive rescaling.
Inner product	$\mathbf{x}^\top \mathbf{y} = r_x r_y \cos \theta$	Yes	No	Combines angular alignment with both radii.
Euclidean distance	$\ \mathbf{x} - \mathbf{y}\ $	Yes	Yes	Sensitive to radial and angular differences.
Angular distance	$\arccos(\cos(\mathbf{x}, \mathbf{y}))$	No	Yes, on $\mathbb{S}^{d-1}$	Geodesic distance between directions.
Chord distance	$\ \mathbf{u}_x - \mathbf{u}_y\ $	No	Yes, on $\mathbb{S}^{d-1}$	Euclidean metric with the same ranking as angular distance.

in the 1960s [11]. An influential later account of the vector-space model represented documents and queries by weighted term vectors and described angular proximity as a comparison rule [12]. In this setting, raw magnitude is strongly influenced by document length and total term count. Angular normalization offered a direct way to compare term distributions without allowing long documents to dominate solely because they contained more tokens. This was an appropriate invariance for many sparse retrieval problems, although it did not remove the need for careful term weighting.

Cosine was not the only retrieval principle, and later probabilistic systems should not be described as naive length-sensitive competitors. BM25, developed after the early vector-space work, explicitly combines term saturation, inverse-document-frequency weighting, and document-length normalization [13]. The historical lesson is therefore not that cosine solved a defect ignored by every alternative. Rather, several retrieval traditions encoded different assumptions about frequency, length, and relevance.

4.2 Static and sentence embeddings

Dense word embeddings strengthened the association between semantic relatedness and local vector geometry. Skip-gram with negative sampling is trained with word–context dot products [14, 15], while GloVe fits a weighted model of log co-occurrence counts [16]. Cosine became a common *evaluation* rule because it suppresses frequency- and scale-related variation and often gives useful neighborhoods. It is important, however, not to equate the downstream cosine rule with the training objective: the latter can depend on both direction and magnitude.

The familiar word-analogy calculation

$$\mathbf{v}_{\text{king}} - \mathbf{v}_{\text{man}} + \mathbf{v}_{\text{woman}}$$

uses cosine or another nearest-neighbor rule to select a candidate, but the relational structure is learned by the embedding model; it is not created by cosine similarity itself. Sentence-BERT later trained siamese encoders to produce sentence vectors intended for efficient cosine-based inference [17]. Its regression variant optimizes cosine directly, whereas its classification and triplet variants use different objectives; train–score alignment is therefore variant-specific. SimCSE made the cosine contract explicit for sentence embeddings by optimizing a contrastive objective over normalized representations [18].

4.3 Contrastive, non-contrastive, and multimodal learning

Modern representation learning gives cosine a more structural role. SimCLR normalizes projection-head outputs and uses a temperature-scaled contrastive objective [1]. MoCo uses a related contrastive formulation with normalized representations [19]. CLIP and ALIGN learn joint image–text spaces with normalized cross-modal scores and a learned or tuned temperature [3, 20]. On the sphere, contrastive learning can be interpreted through alignment of positive pairs and uniformity of the representation distribution [2].

Not every self-supervised method relies on negative examples. BYOL compares normalized predictions and targets while avoiding explicit negative pairs [21]. This matters conceptually: normalization and cosine-based alignment do not uniquely imply a contrastive loss. The training mechanism, stop-gradient operation, predictor architecture, and score must be discussed separately.

Nor is dense retrieval universally cosine-based. Dense Passage Retrieval trains dual encoders with an inner-product score [22]. Across modern systems, cosine is best understood as one common model contract among several, not as the inevitable definition of semantic proximity.

4.4 Why it often works

The historical record suggests four recurring reasons for cosine’s effectiveness:

1. **Useful invariance.** Positive radial scale is frequently a nuisance generated by document length, activation scale, or encoder parameterization.
2. **Objective alignment.** When training normalizes embeddings, angular comparison is part of the learned geometry.
3. **Computational convenience.** Unit vectors reduce cosine search to inner-product search and support efficient nearest-neighbor indexing.
4. **Interpretability.** Direction and angle provide an intuitive, bounded description of alignment.

None of these reasons establishes universal optimality. Each is a hypothesis about the representation and task.

5 What Can an Embedding Norm Mean?

Equation (2) separates radius from direction, but it does not assign semantics to either component. Any interpretation of r_x must be established from the model or from controlled evidence.

5.1 Representative evidence

Table 3 collects the following studies as selected evidence, not as a systematic estimate of how often norms are useful.

For static word embeddings trained with skip-gram negative sampling, Oyama et al. [4] connect squared norm to an information-gain quantity derived from the word’s co-occurrence distribution. They also provide experiments designed to separate that relationship from simple word-frequency correlation. In contextual BERT representations, Zhou et al. [24] show that cosine similarity systematically underestimates similarities involving frequent words relative to human judgments, even after controlling for several confounders. Their geometric explanation is informative but should not be generalized to every contextual encoder without verification.

In self-supervised vision, Draganov et al. [5] show that pre-normalization norms influence cosine-gradient dynamics and, in their studied settings, relate to confidence or sample unexpectedness. These results are strong evidence that a radius discarded by the forward score can still matter during learning.

Amigó et al. [23] extend the norm–information question to contextual and composed text embeddings. Their results show that the correspondence depends strongly on the subword representation and composition rule rather than holding automatically for pooled outputs.

5.2 Why the interpretation is conditional

The same norm can also encode unwanted effects: input length, token frequency, modality-specific scale, pooling choice, layer normalization, or domain shift. Furthermore, a deployed system may expose only unit-normalized outputs, in which case every downstream norm is identically one and the informative pre-normalization radius is unavailable. Evidence about a hidden representation does not automatically justify a norm-aware score on the released representation.

A careful analysis should therefore ask:

- Which layer and which side of the normalization operation supplies the vector?
- Does norm predict a task outcome after controlling for length, frequency, class, and domain?
- Does a norm-aware score improve held-out ranking or calibration, rather than merely correlate in-sample?
- Is the relationship stable across model checkpoints and deployment populations?

The defensible claim is not that norm is always signal. It is that *norm is a model-dependent variable whose removal should be tested when the representation contract does not already declare it irrelevant.*

6 A Taxonomy of Failure Mechanisms

The four operational diagnoses require different evidence and different first responses. Table 4 provides the compact map used in this section.

Table 3: Selected evidence about embedding norms and cosine behavior. Findings are conditional on the listed representation and evaluation setting.

Study	Representation	Reported finding	Qualification
Oyama et al. [4]	Static skip-gram word embeddings	Squared norm is connected to an information-gain quantity and is not explained only by word frequency in the reported experiments.	The derivation and evidence are objective- and model-specific.
Amigó et al. [23]	Contextual and composed text embeddings	Norm-information correspondence changes with the subword representation and text-composition rule.	The correspondence is not automatic for pooled outputs.
Zhou et al. [24]	Contextual BERT word representations	Cosine underestimates human-rated similarities involving high-frequency words in the studied settings.	The result does not establish the same bias for every encoder or layer.
Draganov et al. [5]	Self-supervised vision representations	Pre-normalization norm affects cosine-gradient dynamics and relates to confidence or sample unexpectedness in the studied models.	The evidence concerns pre-normalization, model-dependent quantities.

Table 4: Failure mechanisms, diagnostics, and targeted responses.

Mechanism	Diagnostic	Candidate response	Main caveat
Radial information loss	Norm-outcome association after confounder adjustment; cosine versus dot-product ablation	Inner product, explicit norm features, learned angular-radial score	Norm may be nuisance or unavailable after normalization.
Global geometric distortion	Mean direction, covariance spectrum, pairwise-score spread, neighbor-score margins, layerwise comparison	Centering, component removal, regularized whitening, dimensionality reduction, learned transformation	The remedy is distribution-dependent and may remove useful signal.
Local density and hubness	N_k distribution, skewness, neighbor overlap, cross-modal imbalance	CSLS, querybank normalization, reciprocal or local-density ranking	Requires a representative reference bank and extra computation.
Task-score mismatch	Compare task metrics, symmetry requirements, reliability diagrams, domain shifts	Objective-matched score, bilinear model, calibration, reranker or cross-encoder	More flexible scores require labels, computation, or both.

6.1 Radial information loss

Suppose $\mathbf{x}_1 = r_1 \mathbf{u}$ and $\mathbf{x}_2 = r_2 \mathbf{u}$ with $r_1 \neq r_2$. For every nonzero $\mathbf{y}$,

$$\cos(\mathbf{x}_1, \mathbf{y}) = \cos(\mathbf{x}_2, \mathbf{y}). \quad (11)$$

No downstream procedure that receives only cosine scores can recover the difference between r_1 and r_2 . This is a true limitation precisely when the radius predicts relevance, confidence, informativeness, or another target after nuisance variables are controlled. If radius is arbitrary activation scale, Equation (11) is a benefit instead.

6.2 Global geometric distortion

Global anisotropy is a departure of the directional distribution from the rotationally invariant ideal. For unit vectors $\{\mathbf{u}_i\}_{i=1}^n$ with mean $\bar{\mathbf{u}}$, the average

off-diagonal cosine is

$$A_{\cos} = \frac{1}{n(n-1)} \sum_{i \neq j} \mathbf{u}_i^\top \mathbf{u}_j = \frac{n \|\bar{\mathbf{u}}\|^2 - 1}{n-1}. \quad (12)$$

A large positive value therefore diagnoses a common mean direction. It is not a complete definition of isotropy: a zero-mean distribution can still have a highly nonuniform covariance spectrum. Contextual word representations from several pretrained models have been found to occupy narrow cones [6], and anisotropic sentence spaces can degrade raw similarity comparisons [25].

Anisotropy is not an inherent property of every Transformer. Machina and Mercer [26] identify large Pythia models with near-isotropic representations and connect the result to architecture and final layer normalization. Claims about anisotropy should therefore specify the model, layer, pooling

rule, corpus, and diagnostic.

6.2.1 Distance concentration

In generic high-dimensional settings, distance-based proximity can lose relative contrast [27]. For cosine on the unit sphere, an analogous baseline follows directly from random directions. If U and V are independent and uniform on $\mathbb{S}^{d-1}$, then

$$\mathbb{E}[U^\top V] = 0, \quad \text{Var}(U^\top V) = \frac{1}{d}. \quad (13)$$

Thus pairwise cosines concentrate around zero as dimension grows, while the equivalent chord distances concentrate around $\sqrt{2}$.

Random-pair concentration alone does not show that semantic retrieval has degraded: relevant pairs need not resemble independent uniform directions, and corpus size and signal structure affect the nearest-neighbor tail. Diagnostics should therefore include score margins or relative contrast in the actual embedding corpus, not dimension alone.

6.3 Local density and hubness

Hubness is related to concentration but distinct. Given a query set Q , gallery G , and top- k neighborhood $\mathcal{N}_k(q)$, define the k -occurrence of gallery item g by

$$N_k(g) = \sum_{q \in Q} \mathbf{1}\{g \in \mathcal{N}_k(q)\}. \quad (14)$$

Hubs have unusually large $N_k(g)$, producing a right-skewed occurrence distribution [7]. They can arise with Euclidean or angular neighborhoods and can be amplified when query and gallery distributions differ, as in cross-modal retrieval [8]. Hubness is not simply information contained in norm, so restoring radii is not a general remedy.

6.4 Task and calibration mismatch

Cosine is symmetric:

$$\cos(\mathbf{x}, \mathbf{y}) = \cos(\mathbf{y}, \mathbf{x}).$$

It therefore cannot by itself represent an intrinsically asymmetric relation such as textual entailment, query-to-document utility, or directed substitutability. A bi-encoder can still encode roles through different query and document encoders, but the suitability of a symmetric interaction must be established by training and evaluation.

Cosine is also bounded but uncalibrated. A score of 0.8 is not a probability, and a threshold

learned for one model, layer, language, or corpus need not transfer to another. Ranking quality, classification calibration, and semantic interpretation are different evaluation targets.

7 Targeted Remedies

7.1 Start with the model contract

The first alternative to blind cosine use is not a more elaborate metric; it is to identify the score used during training.

- If the model emits unit vectors and was trained with normalized similarity, cosine, dot product, and Euclidean distance have identical rankings by Equation (9). Choose among them for implementation convenience, then calibrate separately if thresholds matter.
- If the model was trained with an unnormalized inner product, compare inner product and cosine on held-out task metrics before discarding the learned radii.
- If the embeddings were not trained for pairwise comparison, treat every off-the-shelf similarity rule as a baseline rather than a semantic ground truth.

7.2 Centering, component removal, and whitening

When a common direction dominates, mean-centering replaces $\mathbf{x}$ by $\mathbf{x} - \boldsymbol{\mu}$, where $\boldsymbol{\mu}$ is estimated on a reference corpus. All-but-the-top additionally removes a small number of leading principal directions [28]. If the centered covariance is $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, a regularized whitening map has the form

$$T_{\text{white}}(\mathbf{x}) = (\boldsymbol{\Lambda} + \varepsilon \mathbf{I})^{-1/2} \mathbf{U}^\top (\mathbf{x} - \boldsymbol{\mu}), \quad (15)$$

possibly followed by dimensionality reduction and unit normalization. Whitening-based methods have improved unsupervised sentence embeddings in specific BERT settings [29]; normalizing-flow transformations provide a nonlinear alternative [25].

These are estimated transformations, not universal repairs. The reference corpus must represent deployment conditions; covariance estimation may require regularization; and flattening high-variance directions can erase task signal. Every post-processing step should be fitted without test leakage and evaluated against an unprocessed baseline.

7.3 Local-density and querybank corrections

Cross-domain similarity local scaling (CSLS) adjusts a raw score using local neighborhood averages [30]. For cosine or another base score $s(q, g)$, let

$$r_G(q) = \frac{1}{k} \sum_{g' \in \mathcal{N}_G(q)} s(q, g'), \quad (16)$$

$$r_Q(g) = \frac{1}{k} \sum_{q' \in \mathcal{N}_Q(g)} s(q', g). \quad (17)$$

Then

$$s_{\text{CSLS}}(q, g) = 2s(q, g) - r_G(q) - r_Q(g). \quad (18)$$

The gallery-dependent term penalizes items that are broadly similar to many queries.

Querybank normalization generalizes this idea for cross-modal retrieval [8]. Given a bank of representative queries $B = \{b_i\}_{i=1}^m$ and gallery items $\{g_j\}$, precompute the probe matrix $P_{ji} = s(b_i, g_j)$. One option, inverted softmax, uses

$$s_{\text{IS}}(q, g_j) = \frac{\exp\{\beta s(q, g_j)\}}{\sum_{i=1}^m \exp\{\beta P_{ji}\}}, \quad (19)$$

where $\beta > 0$ is an inverse temperature. The denominator depends on the gallery item g_j , so it can change rankings by penalizing candidates activated by many bank queries. By contrast, subtracting one mean and dividing by one standard deviation shared by all candidates for a fixed query cannot change that query’s ranking.

7.4 Norm-aware scoring

The inner product is the simplest norm-aware alternative:

$$s_{\text{dot}}(\mathbf{x}, \mathbf{y}) = r_x r_y \cos(\mathbf{x}, \mathbf{y}). \quad (20)$$

It is appropriate when the product of radii has a validated interpretation, but it can over-rank vectors with large nuisance norms.

When labeled validation data are available, angle and radius can instead enter as separate features. For nonzero query and gallery embeddings, a modest asymmetric score family is

$$s_{\theta}(\mathbf{q}, \mathbf{g}) = \theta_0 + \theta_1 \cos(\mathbf{q}, \mathbf{g}) + \theta_2 \log r_q + \theta_3 \log r_g + \theta_4 |\log r_q - \log r_g|. \quad (21)$$

This equation is a diagnostic model, not a universally recommended similarity. For a fixed query, θ_0 and $\theta_2 \log r_q$ are constant across gallery items and

therefore cannot change within-query ranking; they matter for cross-query thresholds or calibration. A probabilistic decision still requires a fitted link or calibration map. The parameters must be learned on the target task, regularized, and compared with cosine and inner-product baselines. The separated coefficients make the claimed role of each radius empirically falsifiable.

7.5 Word Rotator’s Distance

Word Rotator’s Distance (WRD) is a principled example of using norm and direction for different purposes at the token level [31]. For a sentence with token embeddings $\mathbf{e}_i = \lambda_i \mathbf{u}_i$, where $\lambda_i = \|\mathbf{e}_i\|$ and $\|\mathbf{u}_i\| = 1$, define

$$\nu_s = \sum_i \frac{\lambda_i}{Z_s} \delta_{\mathbf{u}_i}, \quad Z_s = \sum_i \lambda_i. \quad (22)$$

Token norms supply transport mass, while angular dissimilarity supplies the cost

$$c(\mathbf{u}_i, \mathbf{v}_j) = 1 - \mathbf{u}_i^T \mathbf{v}_j.$$

For two sentences with normalized masses $\mathbf{a}$ and $\mathbf{b}$, WRD is the optimal-transport problem

$$\text{WRD}(s, t) = \min_{\Pi \geq 0} \sum_{i,j} \Pi_{ij} c(\mathbf{u}_i, \mathbf{v}_j), \quad (23)$$

$$\text{subject to } \Pi \mathbf{1} = \mathbf{a}, \quad \Pi^T \mathbf{1} = \mathbf{b}.$$

Because its ground cost is cosine dissimilarity, WRD is not guaranteed to satisfy the triangle inequality. The word “distance” here follows the method’s name rather than asserting metric status. WRD is therefore not a radial difference plus one angle between two sentence vectors. It is a distributional alignment method whose usefulness depends on token norms being meaningful importance weights.

7.6 Learned and asymmetric comparison

When a fixed geometric rule cannot express the task, one can learn the interaction. A bilinear score

$$s_M(\mathbf{q}, \mathbf{g}) = \mathbf{q}^T \mathbf{M} \mathbf{g}$$

can model dimension-specific interactions and, when $\mathbf{M}$ is not symmetric, directional relations. A multilayer scorer or cross-encoder is more expressive still because it allows the two inputs to interact before the final score. These approaches trade the indexing efficiency and transparency of cosine for task fit. A common practical compromise is dual-encoder retrieval followed by a learned reranker.

8 A Practical Decision Protocol

No scalar diagnostic selects a universally best comparison rule. The following ordered protocol turns the choice into a reproducible model-selection problem.

1. **Document the representation.** Record model, checkpoint, layer, pooling rule, dimensionality, and whether the exposed vector is before or after normalization.
2. **Identify the training score.** Determine whether the model was optimized with cosine, normalized squared distance, inner product, a classification loss, or no pairwise objective.
3. **Establish equivalent baselines.** Verify vector norms. If they are constant, confirm that cosine, dot product, and Euclidean distance give identical rankings; do not report them as independent methods.
4. **Diagnose the failure.** Measure norm associations, global geometry and score margins, neighborhood hubness, and task calibration separately. Use a representative reference corpus and avoid test leakage.
5. **Apply the smallest targeted remedy.** Test objective-matched scoring before post-processing; post-process for global geometry; use local or querybank corrections for hubness; and learn radial or asymmetric terms only when the task supports them.
6. **Evaluate the actual decision.** For retrieval, report recall, mean reciprocal rank, or nDCG as appropriate. For semantic textual similarity, report correlation with human judgments. For thresholded decisions, report discrimination and calibration. Include domain-shift and subgroup checks when deployment requires them.

This protocol also clarifies what evidence should accompany a proposed alternative. A new score should state its invariances, identify the information it adds or removes, compare against objective-matched baselines, and demonstrate improvement on the downstream decision rather than on an appealing geometric visualization alone.

9 Discussion

9.1 Similarity belongs to the model–task pair

An embedding has no unique semantic geometry in isolation. Its coordinates are the output of an

objective, architecture, data distribution, regularization scheme, and pooling rule. The comparison function adds another inductive bias. Accordingly, semantic similarity should be treated as a task-defined relation—paraphrase, topical relatedness, substitutability, relevance, entailment, or visual correspondence—rather than a universal scalar already present in every vector space.

This perspective explains both the durability and the limitations of cosine. It is durable because positive scale invariance is often useful and because many successful models explicitly train a spherical interface. It is limited because the same invariance can erase information, while angle alone cannot repair global distortion, local-density imbalance, task asymmetry, or miscalibration. The evidence needed for any norm-aware claim is summarized in Section 5; evidence from one model or layer should not be assigned to another by analogy alone.

9.2 Scope and limitations of this review

This article does not attempt a benchmark-by-benchmark meta-analysis or a catalogue of every similarity function. It focuses on the recurring conceptual questions that connect sparse retrieval, representation learning, sentence embeddings, and cross-modal retrieval. Topics such as hyperbolic embeddings, kernel learning, probabilistic embeddings, optimal transport beyond WRD, late-interaction retrieval, and approximate-nearest-neighbor engineering deserve fuller reviews of their own. The selective scope is intended to make the decision logic explicit, not to imply that the listed remedies exhaust the field.

10 Conclusion

Cosine similarity earned its central role for good reasons. It offers a clear geometric interpretation, removes positive radial scale, supports efficient search, and matches the pairwise interaction used in many normalized training objectives. Its radial invariance is best justified when the interface fixes the radius or when norm variation is demonstrably nuisance; the angular geometry and downstream decision still require validation.

Its limitations are equally precise. Cosine cannot use radial information; it may be applied to a geometry not identified by the training task; it does not prevent anisotropy, distance concentration, or hubness; and its symmetric, uncalibrated score may not match the downstream relation. These failures

call for different remedies. Inner products or explicit norm features address radial loss. Centering, component removal, whitening, or learned transformations address global geometry. CSLS and querybank methods address local-density and hubness effects. Learned scorers and rerankers address task-specific or asymmetric relations.

The practical conclusion is therefore not that semantics is always angle, nor that vector norms are always signal. It is that a similarity rule is part of the model. Its invariances should be stated, its geometry diagnosed, and its value judged by the decision the system must make.

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The author used a large language model to assist with literature organization, typographic refinement, and revision of exposition. The author remains responsible for the mathematical statements, source verification, and final content.

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